

COURSE OUTLINE

Course ID	Course Title	Credits
AI205DE01	Trí Tuệ Nhân Tạo	03
	Artificial Intelligence	

(Applied to Semester: 1, Academic year: 2024-2025,
Decision No. 1852/QĐ-DHHS dated 24/7/2024)

A. Course Specifications:

B. Periods				Periods in classroom		
Total periods*	Lecture/Seminar	Activity	Self-study periods	Lecture room	Lab room	Fieldwork
(1)	(2)	(3)	(4)	(5)	(6)	(7)
60	30	30	90	30	30	0

(1) = (2) + (3) = (5) + (6) + (7)

C. Other related Subjects:

Other related Subjects	Course ID	Course title
Prerequisites:		
1.	AI104D	Lập trình cơ bản (Introduction to Programming)

D. Course Description:

Artificial intelligence (AI) is all around us, seamlessly integrated into our daily lives and work. In this course, you will uncover the fundamentals of AI, explore its diverse applications, and understand how it is transforming different industries.

You'll get a general overview of Machine Learning topics such as supervised vs unsupervised learning, and the usage of each algorithm. Also, you understand the advantage of using Python libraries for implementing Machine Learning models. You'll also learn the basics of generative AI and explore its use cases and applications.

Learn and gain a solid foundation in the current techniques of Artificial Intelligence
Grasp the fields where Artificial Intelligence is applied in this era
Apply knowledge of Artificial Intelligence to develop artificial intelligence systems.

E. Course Objectives:

Stt	Course Objectives
O1	To develop a solid foundation in the current techniques of Artificial Intelligence.
O2	To grasp the fields where Artificial Intelligence is applied in this era.
O3	To Apply knowledge of Artificial Intelligence developing artificial intelligence systems.
O4	To help students develop teamwork skills

F. Learning Outcomes:

No.		Learning Outcomes	PLOs
O1	CLO1.	Understand the principles of problem solving in artificial intelligence and be able to successfully apply a number of AI methods.	PLO 2
O3	CLO2.	Implement many popular AI algorithms using the Python programming language to solve business problems and identify opportunities.	PLO 3
O2	CLO3.	Understand some real-life AI applications and topics such as machine learning, deep learning, generative artificial intelligence, etc.	PLO 2
O2	CLO4.	Seriously discuss the positive and negative impacts of AI on individuals, organizations and society	PLO 1
O4	CLO5.	Participate and work effectively in groups.	PLO 5

G. Instructional Modes:

	Room	Periods
1	Lecture room	30
2	Lab room	30
	Total	60

Lectures will be combined with in-class discussion and group presentation with some requirements: students must read book, do homework, practice and prepare lesson before going to the class.

Lecture in class

1. The time for lecturing is 30 theoretical hours (50% of the duration of the module) taking place in 2 periods/session and in 15 sessions (if it is a main semester, then 1 session/week, if it is an extra semester 2 sessions/week).
2. Students read textbooks and lectures in English, lecture in Vietnamese with English annotations for terms.
3. Before going to class, students read the textbook chapters specified in the syllabus at home.
4. Students will come to class to watch and hear the instructor highlight the important or difficult concepts and ideas of each chapter.
5. After the lecture, students make their own examples given in the lecture and/or in the book to fully understand the concepts.
6. If students have problems with steps 2-4 above, students need to discuss them with classmates or bring them to practice time to discuss.

Presentation

1. Students present technology topics during theory classes in class according to the instructor's arrangement.
2. Students will be divided into small groups of no more than 3 students to carry out the topic under the guidance of instructors.
3. Students can discuss with lecturers and groups about difficulties in the process of understanding the topic.

Exercises And Practice

1. Thirty hours for exercises and practice are arranged interspersed with theory hours in class, taking place in 15 sessions.
2. Students do exercises as requested by the instructor to test their mastery of knowledge.

3. Before homework time, students should try as many of the given exercises as possible.
4. Record the difficulties when trying to do this exercise first, then bring these difficulties to ask or discuss during exercise time.
5. In each exercise session, the lecturer will guide students to complete the weekly exercises.
6. Teachers will require assignments to be submitted and graded several weeks into the course.

H. Textbooks and teaching aids:

1. Textbooks:
 - [1] Alberto Artasanchez, Prateek Joshi, *Artificial Intelligence with Python*, Second Edition, Packt Publishing, 2020
 - [2] Patrick D. Smith, *Hands-On Artificial Intelligence for Beginners: An introduction to AI concepts, algorithms, and their implementation*, Packt Publishing, 2018
2. References:
 - [3] Dipanjan Sarkar, Raghav Bali, Tushar Sharma, *Practical Machine Learning with Python: A Problem-Solver's Guide to Building Real-World Intelligent Systems*, Apress, 2018.
 - [4] Tom Taulli, *Artificial Intelligence Basics: A Non-Technical Introduction*, Apress, 2019.
3. Software/Apps:
 - [5] Jupyter. Download tại: <https://jupyter.org/>
 - [6] Anaconda. Download tại: <https://www.anaconda.com/>
 - [7] PyCharm. Download tại: <https://www.jetbrains.com/pycharm/>

I. Assessment Methods (Requirements for Completion of the Course):

1. Description of learning outcomes assessment

a) In-class test: This part accounts for 20% of the total score of the module. Students will be tested on their knowledge of the previous week through multiple choice questions, short questions and practice exercises.

b) Case study: This section accounts for 30% of the total score of the module, determined and implemented by the teacher.

c) Final project:

This part accounts for 50% of the total score of the module. Students carry out the topic according to the teacher's request, working in groups of maximum 3 students. Submit your assignment at the last class session.

2. Summary of learning outcomes assessment

* For main/extra semester:

Components	Duration	Assessment Forms	Percentage	Schedule	CLOs
In-class test	10'	Multiple choice tests or short questions during theory time and practice exercises in class. The number of assessments is determined by the teacher.	20%	Week 2-14	1-4
Case study		Work in groups of maximum 3 students. Group presentation in class.	30%	Week 2-14	3-5
Final Project		Work in groups of maximum 3 students.	50%	Week 15	1-5

		<i>Topics are determined by the teacher.</i>			
Total			100%		

I. Academic integrity

Academic integrity is a fundamental value that affects the quality of teaching, learning, and research at a university. To ensure the maintenance of academic integrity at Hoa Sen University, students are required to:

1. Work independently on individual assignments: Collaborating on individual assignments is considered cheating.
2. Avoid plagiarism: Plagiarism is an act of fraud that involves the use of ideas or words of another person without proper attribution. Students will be accused of plagiarism if they:
 - a. Copy in their work one or more sentences from another person without proper citation.
 - b. Rephrase, paraphrase, or translate another person's ideas or words without proper attribution.
 - c. Reuse their own assignments, in whole or in part, and submit them for another class.
 - d. Work responsibly within a working group.
3. In cooperative group assignments, all students are required to stay on task and contribute equally to the projects. Group reports should clearly state the contribution of each group member. Any acts of academic dishonesty will result in a grade of zero for the task at hand and/or immediate failure of the course, depending on the seriousness of the fraud. Please consult Hoa Sen University's Policy on Plagiarism at <http://thuvien.hoasen.edu.vn/chinh-sach-phong-tranh-dao-van>

To ensure the maintenance of academic integrity, the university asks that students report cases of academic dishonesty to the teacher and/or the Dean. The names of those students will be kept anonymous.

J. Teaching Staff:

No.	Lecturer's name	Email, Phone number, Office location	Office hours	Position
1	Ts. Lê Đình Phong	phong.ledinh@hoasen.edu.vn		Lecturer

K. Outline of Topics to be covered:

- For main/extra semester:

Week	Lecture Content	Reference Material	Work to be completed by students	CLOs
1/1	Introduction to AI • What is AI? • Impact of AI, Applications and Examples • Branches of AI • What Can AI Do? • History of AI	[1]. Chapter 1,2	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 1

	Fundamental Use Cases for Artificial Intelligence			
2/2	Machine Learning Pipelines - What is a machine learning pipeline? - Problem definition - Data ingestion - Data preparation - Data segregation - Model training	[1]. Chapter 3	- Read the materials before going to theory class. - Complete assignments in practical class and submit to Mlearning	CLO 1,3
3/3	Feature Selection and Feature Engineering - Feature selection - Feature engineering - Imputation - Outlier management - One-hot encoding - Log transform - Scaling - Date manipulation	[1]. Chapter 4	- Read the materials before going to theory class. - Complete assignments in practical class and submit to Mlearning	CLO 1,3
4/4	Classification and Regression Using Supervised Learning - Supervised versus unsupervised learning - What is classification? - Preprocessing data - Label encoding - Logistic regression classifiers - The Naïve Bayes classifier - Confusion matrixes - Support Vector Machines - Classifying income data using Support Vector Machines - What is regression? - Building a single-variable regressor - Building a multivariable regressor	[1]. Chapter 5	- Read the materials before going to theory class. - Complete assignments in practical class and submit to Mlearning	CLO 2
5/5	Predictive Analytics with Ensemble Learning - What are decision trees? - Building a decision tree classifier - What is ensemble learning? - Building learning models with ensemble learning - What are random forests and extremely random forests? - Building random forest and extremely random forest classifiers - Estimating the confidence measure of the predictions - Dealing with class imbalance - Finding optimal training parameters using grid search	[1]. Chapter 6	- Read the materials before going to theory class. - Complete assignments in practical class and submit to Mlearning	CLO 2

	• Computing relative feature importance			
6/6	Detecting Patterns with Unsupervised Learning • What is unsupervised learning? • Clustering data with the K-Means algorithm • What are Gaussian Mixture Models? • Finding subgroups in stock market using the Affinity • Propagation model • Segmenting the market based on shopping patterns	[2]. Chapter 7	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
7/7	Building Recommender Systems • Extracting the nearest neighbors • Building a K-nearest neighbors classifier • Computing similarity scores • Finding similar users using collaborative filtering • Building a movie recommendation system	[1]. Chapter 8	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
8/8	Sequential Data and Time Series Analysis • Understanding sequential data • Handling time series data with Pandas • Slicing time series data • Operating on time series data • Extracting statistics from time series data • Generating data using Hidden Markov Models • Identifying alphabet sequences with Conditional Random Fields • Stock market analysis	[1]. Chapter 17	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
9/9	Neural Networks • Introduction to neural networks • Building a Perceptron-based classifier • Constructing a single-layer neural network • Constructing a multi-layer neural network • Building a vector quantizer • Analyzing sequential data using recurrent neural networks • Visualizing characters in an optical character recognition database	[1]. Chapter 19	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2

	• Building an optical character recognition engine			
10/10	Deep Learning with Convolutional Neural Networks • The basics of Convolutional Neural Networks • Architecture of CNNs • Types of layers in a CNN • Building a perceptron-based linear regressor • Building an image classifier using a single-layer neural network • Building an image classifier using a Convolutional Neural Network	[1]. Chapter 20	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
11/11	Recurrent Neural Networks and Other Deep Learning Models • The basics of Recurrent Neural Networks • Architecture of RNNs • A language modeling use case • Training an RNN	[1]. Chapter 21	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
12/12	Creating Intelligent Agents with Reinforcement Learning • Understanding what it means to learn • Reinforcement learning versus supervised learning • Real-world examples of reinforcement learning • Building blocks of reinforcement learning • Creating an environment • Building a learning agent	[1]. Chapter 22	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
13/13	Artificial Intelligence and Big Data • Big data basics • The three V's of big data • Big data and machine learning • NoSQL Databases	[1]. Chapter 23	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
14/14	AI: Issues, Concerns and Ethical Considerations 1. Exploring Today's AI Concerns 2. Exploring AI and Ethics 3. Defining AI Ethics 4. Understanding Bias and AI 5. AI Ethics and Regulations 6. AI Ethics, Governance, and ESG Our future with AI 1. The evolution and future of AI	[4]. Chapter 9	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 4

	2. Future with AI 3. What will our Society look like when AI is everywhere? Your future with AI 1. The AI Ladder - The Journey for Adopting AI Successfully 2. Advice for a career in AI 3. Hotbeds of AI Innovation			
15/15	Final Project Grading		Complete all project tasks and submit to Mlearning	CLO 1-5

Date: 12/7/2024
Written by
(Signature and the full name)



Lê Thanh Tùng

Date: 12/7/2024
Programme Chair
(Signature and the full name)



Lê Thanh Tùng

Date: 12/7/2024
Dean/ Vice Dean
(Signature and the full name)



Lê Đình Phong